

Alper Sahistan,

Ph.D. Student

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Research Interests

Ph.D. student with expertise in *ray tracing*, *volume rendering*, *scientific visualization*, *computer graphics*, and *data compression*. I am also interested in high-performance computing and computational geometry.

Education

University of Utah, Scientific Computing and Imaging (SCI) Institute

Ph.D Student

2022–current

Bilkent University Department of Computer Engineering

M.Sc. in Computer Science

2019–2022

Bilkent University Department of Computer Engineering

B.S. in Computer Science

2015–2019

Research Experience

University of Utah

- Pursuing a Ph.D. under the advisement of Prof. Valerio Pascucci, focusing on scientific visualization, including large-scale volume rendering, multi-field representations, and data compression.
- Collaborating with Dr. Ingo Wald (NVIDIA), Dr. Stefan Zellmann (Univ. of Cologne), and Nate Morrical (NVIDIA) on GPU ray tracing, Woodcock tracking, and data-parallel rendering projects.
- Contributing to the **Intel Center of Excellence** at the University of Utah and working with **Lawrence Livermore National Laboratory** to implement the DPC++/SYCL version of **zfp**, an open-source, error-bounded compression library for multidimensional scientific data.

Bilkent University

- Completed an M.Sc. under the advisement of Prof. Uğur Güdükbay, with a research focus on ray tracing and scientific visualization techniques.
- Developed a data-parallel visualization framework capable of rendering and compositing non-convex unstructured meshes with elements such as tetrahedra, wedges, and pyramids using XOR-compacted data structures.
- Researched acceleration structures for ray tracing by leveraging cache-efficient, bitwise XOR-compacted tetrahedral meshes, and 2D projections to optimize traversal.
- The research was supported by **The Scientific and Technological Research Council of Turkey (TÜBİTAK)**.

Industry Experience

NASA Ames Research Center

Co-op Research Intern
(via Analytical Mechanics Assoc.)

Developed the rendering backend for Discover-X, a data-parallel visualization tool for large-scale scientific data with browser-based, real-time interaction via WebSockets, built on NVIDIA Barney, ANARI, and related GPU APIs.

2026

Lawrence Livermore National Laboratory

Graduate Computing Intern

Explored vector field compression via Gaussian mixture models which evaluated loss over Clebsch or Helmholtz representations.

2025

Lawrence Livermore National Laboratory

Graduate Computing Intern

Worked on profiling and optimizing floating-point compression library **zfp**. Started porting **zfp** to SYCL.

2023

TaleWorlds Entertainment

Engine Programming Intern

Worked at the internal 3D rendering engine of strategy/action RPG video game *Mount&Blade II: Bannerlord*

2018

Presentations

1. A. Gooch, A. Panta, A. Sahistan, X. Huang, M. Taufer, J. Marquez, G. Scorzelli, and V. Pascucci (2024). Enabling Scientific Discovery: A Tutorial for Harnessing the Power of the National Science Data Fabric for Large-Scale Data Analysis. In: *2024 IEEE Visualization Conference (VIS) Tutorials*.
2. A. Sahistan, N. Morrical, P. Lindstrom, and V. Pascucci (2023). Building a portable, scalable, performant ZFP backend using oneAPI and SYCL to advance exascale computing: a developer perspective. In: *oneAPI DevSummit for AI and HPC 2023*.

Teaching Experience

Bilkent University
Teaching Assistant

Grading assignments for Programming Languages course
taught by Dr. H. Altay Güvenir.
Tutoring and grading Computer Organisation course labs
by Dr. Özcan Öztürk

2019–2022